At the opposite extreme of the spectrum, modern totalitarian regimes like Stalin's USSR believed that humans are capable of almost limitless change. Through relentless social control even deep-seated biological characteristics such as egotism and familial attachments could be uprooted, and a new socialist human created.

Surveillance by state agents, priests and neighbours was key for imposing on people both rigid caste systems and totalitarian reeducation campaigns. New surveillance technology, especially when coupled with a social credit system, might force people either to conform to a novel caste system or to constantly change their actions, thoughts and personality in accordance with the latest instructions from above.

Democratic societies that employ powerful surveillance technology therefore need to beware of the extremes of both over-rigidity and over-pliability. Consider, for example, a national health-care system that deploys algorithms to monitor my health. At one extreme, the system could take an overly rigid approach and ask its algorithm to predict what illnesses I am likely to suffer from. The algorithm then goes over my genetic data, my medical file, my social media activities, my diet and my daily schedule and concludes that I have a 91 per cent chance of suffering a heart attack at the age of fifty. If this rigid medical algorithm is used by my insurance company, it may prompt the insurer to raise my premium.[5] If it is used by my bankers, it may cause them to refuse me a loan. If it is used by potential spouses, they may decide not to marry me.

But it is a mistake to think that the rigid algorithm has really discovered the truth about me. The human body is not a fixed block of matter but a complex organic system that is constantly growing, decaying and adapting. Our minds too are in constant flux. Thoughts, emotions and sensations pop up, flare for a while and die down. In our brains, new synapses form within hours.[6] Just reading this paragraph, for example, is changing your brain structure a little, encouraging neurons to make new connections or abandon old links. You are already a little different from what you were when you began reading it. Even at the genetic level things are surprisingly flexible. Though an individual's DNA remains the same throughout life, epigenetic and environmental factors can significantly alter how the same genes express themselves.

So an alternative health-care system may instruct its algorithm not to *predict* my illnesses, but rather to help me avoid them. Such a dynamic